Two-Way Frequency Tables & Conditional Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If one adult is selected at random from the group, what is the probability that the adult is in the category Southside and Support?

Answer: 5/48. There are 10 such adults out of 96 total, so the probability is $\frac{10}{96} = \frac{5}{48}$.

2 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. If one student is selected at random from the group, what is the probability that the student is in the category Grade 11 and Morning?

Answer: 9/106. There are 9 such students out of 106 total, so the probability is $\frac{9}{106} = \frac{9}{106}$.

3 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If one book is selected at random from the group, what is the probability that the book is in the category Fiction and Digital?

Answer: 11/26. There are 55 such books out of 130 total, so the probability is $\frac{55}{130} = \frac{11}{26}$.

4 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If one commuter is selected at random from the group, what is the probability that the commuter is in the category Under 30 and Car?

Answer: 15/64. There are 30 such commuters out of 128 total, so the probability is $\frac{30}{128} = \frac{15}{64}$.

5 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey of 178 employees found that 12 are in the Marketing department and working remotely, 118 are in the Marketing department and working on-site, 35 are not in the Marketing department and working remotely, and 13 are not in the Marketing department and working on-site. What is the probability that a randomly selected employee is in the Marketing department?

Answer: 65/89. $P(\text{in the Marketing department}) = \frac{130}{178} = 65/89$

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6 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bag contains 5 red marbles and 15 blue marbles. If one marble is drawn at random, what is the probability that it is red?

Answer: 1/4. Applying Probability, Conditional Probability & Two-Way Tables principles, $P(\text{red}) = \frac{5}{20} = 1/4$ Therefore, the verified correct option is 1/4.

7 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey of 178 shoppers found that 47 are using a coupon and spending over USD 50, 75 are using a coupon and spending USD 50 or less, 25 are not using a coupon and spending over USD 50, and 31 are not using a coupon and spending USD 50 or less. What is the probability that a randomly selected shopper is using a coupon?

Answer: 61/89. $P(\text{using a coupon}) = \frac{122}{178} = 61/89$

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If a patient in the category Treatment is selected at random, what is the probability that the patient is also in the category Improved?

Answer: 11/49. Restrict to the 49 patients in Treatment; 11 of them are Improved: $\frac{11}{49} = \frac{11}{49}$.

9 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If a book in the category Nonfiction is selected at random, what is the probability that the book is also in the category Print?

Answer: 23/81. Restrict to the 81 books in Nonfiction; 23 of them are Print: $\frac{23}{81} = \frac{23}{81}$.

10 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Northside is selected at random, what is the probability that the adult is also in the category Support?

Answer: 25/41. Restrict to the 82 adults in Northside; 50 of them are Support: $\frac{50}{82} = \frac{25}{41}$.

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11 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If a book in the category Fiction is selected at random, what is the probability that the book is also in the category Print?

Answer: 21/76. Restrict to the 76 books in Fiction; 21 of them are Print: $\frac{21}{76} = \frac{21}{76}$.

12 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If a frog in the category Pond A is selected at random, what is the probability that the frog is also in the category Brown?

Answer: 17/56. Restrict to the 56 frogs in Pond A; 17 of them are Brown: $\frac{17}{56} = \frac{17}{56}$.

13 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If a patient in the category Placebo is selected at random, what is the probability that the patient is also in the category Did not improve?

Answer: 22/39. Restrict to the 39 patients in Placebo; 22 of them are Did not improve: $\frac{22}{39} = \frac{22}{39}$.

14 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey of 207 voters found that 81 are under 40 years old and supporting the proposal, 5 are under 40 years old and opposing the proposal, 57 are 40 years old or older and supporting the proposal, and 64 are 40 years old or older and opposing the proposal. Given that a voter is under 40 years old, what is the probability that they are also supporting the proposal?

Answer: 81/86. $P(\text{supporting the proposal} \mid \text{under 40 years old}) = \frac{81}{86} = 81/86$

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. A student will be selected at random from those in the category Evening. What is the probability that the student is in the category Grade 11?

Answer: 16/29. There are 58 students in Evening, of whom 32 are in Grade 11: $\frac{32}{58} = \frac{16}{29}$.

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16 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. A frog will be selected at random from those in the category Brown. What is the probability that the frog is in the category Pond B?

Answer: $4/7$. There are 77 frogs in Brown, of whom 44 are in Pond B: $\frac{44}{77} = \frac{4}{7}$.

17 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. One entry in the table is missing. If a patient in the category Placebo is selected at random, what is the probability that the patient is in the category Improved?

Answer: $29/48$. The missing entry is $96 - 38 = 58$. So the probability is $\frac{58}{96} = \frac{29}{48}$.

18 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. A patient will be selected at random from those in the category Improved. What is the probability that the patient is in the category Treatment?

Answer: $3/5$. There are 80 patients in Improved, of whom 48 are in Treatment: $\frac{48}{80} = \frac{3}{5}$.

19 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. One entry in the table is missing. If a adult in the category Southside is selected at random, what is the probability that the adult is in the category Oppose?

Answer: $41/69$. The missing entry is $69 - 28 = 41$. So the probability is $\frac{41}{69} = \frac{41}{69}$.

20 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. A adult will be selected at random from those in the category Support. What is the probability that the adult is in the category Southside?

Answer: $24/35$. There are 70 adults in Support, of whom 48 are in Southside: $\frac{48}{70} = \frac{24}{35}$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If one commuter is selected at random from the group, what is the probability that the commuter is in the category Under 30 and Public transit?

a. $\frac{5}{29}$ ✓

b. $\frac{5}{12}$

c. $\frac{25}{77}$

d. $\frac{12}{29}$

There are 25 such commuters out of 145 total, so the probability is $\frac{25}{145} = \frac{5}{29}$.

22 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If one frog is selected at random from the group, what is the probability that the frog is in the category Pond A and Brown?

a. $\frac{25}{38}$

b. $\frac{10}{19}$

c. $\frac{76}{151}$

d. $\frac{50}{151}$ ✓

There are 50 such frogs out of 151 total, so the probability is $\frac{50}{151} = \frac{50}{151}$.

23 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If one frog is selected at random from the group, what is the probability that the frog is in the category Pond B and Brown?

a. $\frac{65}{146}$

b. $\frac{9}{73}$ ✓

c. $\frac{18}{65}$

d. $\frac{18}{61}$

There are 18 such frogs out of 146 total, so the probability is $\frac{18}{146} = \frac{9}{73}$.

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24 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If one patient is selected at random from the group, what is the probability that the patient is in the category Treatment and Did not improve?

a. $\frac{10}{57}$

b. $\frac{10}{49}$

c. $\frac{57}{140}$

d. $\frac{1}{14}$ ✓

There are 10 such patients out of 140 total, so the probability is $\frac{10}{140} = \frac{1}{14}$.

25 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. If one student is selected at random from the group, what is the probability that the student is in the category Grade 11 and Morning?

a. $\frac{48}{151}$ ✓

b. $\frac{12}{23}$

c. $\frac{24}{29}$

d. $\frac{92}{151}$

There are 48 such students out of 151 total, so the probability is $\frac{48}{151} = \frac{48}{151}$.

26 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If one frog is selected at random from the group, what is the probability that the frog is in the category Pond A and Green?

a. $\frac{18}{37}$

b. $\frac{46}{111}$ ✓

c. $\frac{23}{27}$

d. $\frac{23}{38}$

There are 46 such frogs out of 111 total, so the probability is $\frac{46}{111} = \frac{46}{111}$.

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27 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If one patient is selected at random from the group, what is the probability that the patient is in the category Placebo and Improved?

a. $\frac{83}{175}$

b. $\frac{36}{175}$ ✓

c. $\frac{36}{83}$

d. $\frac{36}{77}$

There are 36 such patients out of 175 total, so the probability is $\frac{36}{175} = \frac{36}{175}$.

28 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If one book is selected at random from the group, what is the probability that the book is in the category Nonfiction and Digital?

a. $\frac{2}{7}$

b. $\frac{6}{13}$

c. $\frac{21}{52}$

d. $\frac{3}{26}$ ✓

There are 12 such books out of 104 total, so the probability is $\frac{12}{104} = \frac{3}{26}$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If a commuter in the category 30 or older is selected at random, what is the probability that the commuter is also in the category Car?

a. $\frac{34}{61}$

b. $\frac{27}{61}$ ✓

c. $\frac{27}{151}$

d. $\frac{27}{68}$

Restrict to the 61 commuters in 30 or older; 27 of them are Car: $\frac{27}{61} = \frac{27}{61}$.

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30 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Southside is selected at random, what is the probability that the adult is also in the category Oppose?

a. $\frac{59}{202}$

b. $\frac{59}{107}$

c. $\frac{55}{114}$

d. $\frac{59}{114}$ ✓

Restrict to the 114 adults in Southside; 59 of them are Oppose: $\frac{59}{114} = \frac{59}{114}$.

31 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Northside is selected at random, what is the probability that the adult is also in the category Oppose?

a. $\frac{17}{27}$ ✓

b. $\frac{34}{127}$

c. $\frac{17}{40}$

d. $\frac{10}{27}$

Restrict to the 54 adults in Northside; 34 of them are Oppose: $\frac{34}{54} = \frac{17}{27}$.

32 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. If a student in the category Grade 11 is selected at random, what is the probability that the student is also in the category Morning?

a. $\frac{3}{4}$

b. $\frac{13}{40}$

c. $\frac{27}{40}$ ✓

d. $\frac{9}{28}$

Restrict to the 40 students in Grade 11; 27 of them are Morning: $\frac{27}{40} = \frac{27}{40}$.

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33 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If a book in the category Fiction is selected at random, what is the probability that the book is also in the category Print?

a. $\frac{9}{26}$ ✓

b. $\frac{9}{109}$

c. $\frac{9}{38}$

d. $\frac{17}{26}$

Restrict to the 26 books in Fiction; 9 of them are Print: $\frac{9}{26} = \frac{9}{26}$.

34 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If a commuter in the category Under 30 is selected at random, what is the probability that the commuter is also in the category Public transit?

a. $\frac{33}{146}$

b. $\frac{33}{82}$

c. $\frac{10}{21}$

d. $\frac{11}{21}$ ✓

Restrict to the 63 commuters in Under 30; 33 of them are Public transit: $\frac{33}{63} = \frac{11}{21}$.

35 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Southside is selected at random, what is the probability that the adult is also in the category Oppose?

a. $\frac{16}{43}$

b. $\frac{24}{47}$

c. $\frac{23}{71}$

d. $\frac{48}{71}$ ✓

Restrict to the 71 adults in Southside; 48 of them are Oppose: $\frac{48}{71} = \frac{48}{71}$.

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36 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If a frog in the category Pond B is selected at random, what is the probability that the frog is also in the category Brown?

a. $\frac{11}{78}$

b. $\frac{11}{53}$

c. $\frac{15}{26}$

d. $\frac{11}{26}$ ✓

Restrict to the 26 frogs in Pond B; 11 of them are Brown: $\frac{11}{26} = \frac{11}{26}$.

37 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If a patient in the category Placebo is selected at random, what is the probability that the patient is also in the category Did not improve?

a. $\frac{22}{95}$

b. $\frac{11}{27}$

c. $\frac{17}{39}$

d. $\frac{22}{39}$ ✓

Restrict to the 39 patients in Placebo; 22 of them are Did not improve: $\frac{22}{39} = \frac{22}{39}$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. One entry in the table is missing. If a student in the category Grade 10 is selected at random, what is the probability that the student is in the category Evening?

a. $\frac{44}{93}$

b. $\frac{49}{150}$

c. $\frac{49}{92}$

d. $\frac{49}{93}$ ✓

The missing entry is $93 - 44 = 49$. So the probability is $\frac{49}{93} = \frac{49}{93}$.

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39 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. A student will be selected at random from those in the category Morning. What is the probability that the student is in the category Grade 10?

a. $\frac{36}{71}$

b. $\frac{12}{47}$

c. 1

d. $\frac{1}{2}$ ✓

There are 72 students in Morning, of whom 36 are in Grade 10: $\frac{36}{72} = \frac{1}{2}$.

40 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. For which group is the proportion in the category Green greater?

a. **Pond A** ✓

b. Pond B

c. The proportions are equal.

d. There is not enough information to compare.

Pond A: $\frac{48}{103} \approx 0.466$; Pond B: $\frac{26}{71} \approx 0.366$. The larger proportion belongs to Pond A.

41 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. One entry in the table is missing. If a frog in the category Pond B is selected at random, what is the probability that the frog is in the category Green?

a. $\frac{23}{62}$

b. $\frac{39}{124}$

c. $\frac{13}{29}$

d. $\frac{39}{62}$ ✓

The missing entry is $62 - 23 = 39$. So the probability is $\frac{39}{62} = \frac{39}{62}$.

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42 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. For which group is the proportion in the category Print greater?

a. Fiction ✓

b. Nonfiction

c. The proportions are equal.

d. There is not enough information to compare.

Fiction: $\frac{53}{82} \approx 0.646$; Nonfiction: $\frac{10}{37} \approx 0.270$. The larger proportion belongs to Fiction.

43 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. One entry in the table is missing. If a book in the category Nonfiction is selected at random, what is the probability that the book is in the category Digital?

a. $\frac{50}{101}$ ✓

b. $\frac{51}{101}$

c. $\frac{25}{86}$

d. $\frac{25}{38}$

The missing entry is $101 - 51 = 50$. So the probability is $\frac{50}{101} = \frac{50}{101}$.

44 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. For which group is the proportion in the category Morning greater?

a. Grade 11

b. The proportions are equal.

c. There is not enough information to compare.

d. Grade 10 ✓

Grade 10: $\frac{35}{69} \approx 0.507$; Grade 11: $\frac{27}{83} \approx 0.325$. The larger proportion belongs to Grade 10.

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45 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. For which group is the proportion in the category Support greater?

a. There is not enough information to compare.

b. **Southside ✓**

c. Northside

d. The proportions are equal.

Northside: $\frac{30}{82} \approx 0.366$; Southside: $\frac{39}{93} \approx 0.419$. The larger proportion belongs to Southside.

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